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Tutorial - 12

① $E_{n_x, n_y} = \frac{\hbar\omega}{2}(n_x + n_y + 1)$

1st excited state: $n_x = 1 ; n_y = 0$, $n_x = 0 ; n_y = 1$] 2 fold degenerate $|1,0\rangle$ & $|0,1\rangle$.

$$x = \sqrt{\frac{\hbar}{2m\omega}}(a_x + a_x^\dagger) \; ; \; y = \sqrt{\frac{\hbar}{2m\omega}}(a_y + a_y^\dagger)$$

$$xy = \frac{\hbar}{2m\omega}(a_x + a_x^\dagger)(a_y + a_y^\dagger)$$

$$H' = c\,xy.$$

We work in the basis of the 2 unperturbed degenerate 1st excited states. $|1,0\rangle$ & $|0,1\rangle$.

$$H' = c\begin{pmatrix} \langle 1,0|xy|1,0\rangle & \langle 1,0|xy|0,1\rangle \\ \langle 0,1|xy|1,0\rangle & \langle 0,1|xy|0,1\rangle \end{pmatrix}$$

Now, $\langle n_x, n_y|xy|n_x, n_y\rangle = \langle n_x|x|n_x\rangle\langle n_y|y|n_y\rangle$
$= \langle n\rangle$.

The diagonal elements vanish as $\langle n|a + a^\dagger|n\rangle = 0$ for $n = 0, 1$.

Off-diagonal: $\langle 1,0|xy|0,1\rangle = \langle 1|x|0\rangle\langle 0|y|1\rangle$.
$= \frac{\hbar}{2m\omega}\langle 1|1\rangle.\langle 0|0\rangle = \frac{\hbar}{2m\omega}$

$= \langle 0,1|xy|1,0\rangle$

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$$\therefore \quad H' = c\begin{pmatrix} 0 & \frac{\hbar}{2m\omega} \\ \frac{\hbar}{2m\omega} & 0 \end{pmatrix} = c\begin{pmatrix} 0 & v \\ v & 0 \end{pmatrix} \quad \left[v = \frac{c\hbar}{2m\omega}\right]$$

Let its eigenvalues be c.

$$\therefore \quad \det\begin{pmatrix} -c & v \\ v & -c \end{pmatrix} = 0 \quad \Rightarrow \quad c^2 - v^2 = 0.$$

$$\Rightarrow \quad \boxed{c = \pm v}$$

NOT REQUIRED

$\underline{c = v}$: $\quad |\psi_+\rangle = A_+\begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix}.$

$$\begin{pmatrix} -v & v \\ v & -v \end{pmatrix}\begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} = 0 \quad \Rightarrow \quad \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} = 0$$

$$\Rightarrow \quad \boxed{\psi_1 = \psi_2}$$

$$\therefore \quad |\psi_+\rangle = A_+\begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$\therefore$ From normalisat^n: $\quad |\psi_+\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}.$

$\underline{c = -v}$: $\quad |\psi_-\rangle = A_-\begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix}.$

$$\therefore \quad \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} = 0 \quad \Rightarrow \quad \boxed{\psi_1 = -\psi_2}$$

$$\therefore \quad |\psi_-\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

$$\therefore \quad \Delta E = 2v = \frac{2c\hbar}{2m\omega} = \frac{c\hbar}{m\omega} \quad \underline{\text{(Ans)}}$$

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(2) The ground states for this system are: $|0, \uparrow\rangle$, $|0, \downarrow\rangle$.

$E_0 = \frac{\hbar\omega}{2}$; $H_p = \lambda x^2 \sigma_x$.

$\therefore E_0^1 = \langle 0, s | H_p | 0, s' \rangle$.

$= \lambda \langle 0 | x^2 | 0 \rangle \langle s | \sigma_x | s' \rangle$; $s, s' = \uparrow, \downarrow$.

$x^2 = \frac{\hbar}{2m\omega} (a + a^\dagger)^2 = \frac{\hbar}{2m\omega} (a^{\dagger 2} + a^2 + aa^\dagger + a^\dagger a)$

Only $aa^\dagger$ & $a^\dagger a$ will contribute as others will produce orthogonal states.

$\therefore \langle 0 | x^2 | 0 \rangle = \frac{\hbar}{2m\omega} \langle 0 | aa^\dagger + a^\dagger a | 0 \rangle$.

$= \frac{\hbar}{2m\omega} \langle 0 | aa^\dagger | 0 \rangle = \frac{\hbar}{2m\omega}$

$\therefore H_p \rightarrow \langle s | \sigma_x | s' \rangle \longrightarrow$ Pauli x-matrix.

$\therefore H_p = \lambda \frac{\hbar}{2m\omega} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

$\therefore E_0^1 = \pm \frac{\lambda\hbar}{2m\omega}$ (eigenvalues of H_p)

$\therefore E = E_0 + E_0^1 = \frac{\hbar\omega}{2} \pm \frac{\lambda\hbar}{2m\omega}$ (Ans)

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(4) Since $\hat{H}_0$ is diagonal, the exact eigenvalues & eigenenergies are as follows:

(i) ~~Eigenenerg~~ $E_1 : 15E_0$

$|\psi_1\rangle : (1\ 0\ 0\ 0)^T = e_1$

(ii) $E_2 : 3E_0$

$|\psi_2\rangle : (0\ 1\ 0\ 0)^T = e_2$

(iii) $E_3 : 3E_0$

$|\psi_3\rangle : (0\ 0\ 1\ 0)^T = e_3$

(iv) $E_4 : 3E_0$

$|\psi_4\rangle : (0\ 0\ 0\ 1)^T = e_4$

$$H = H_0 + H_P = E_0 \begin{pmatrix} 15 & 0 & 0 & 0 \\ 0 & 3+\frac{3}{100} & \frac{1}{100} & 0 \\ 0 & \frac{1}{100} & 3 & 0 \\ 0 & 0 & 0 & 3+\frac{3}{100} \end{pmatrix}$$

* The 1st eigenstate (e_1) & the 4th eigenstate (e_4) are uncoupled to the other states in the new Hamiltonian H. Hence, they remain eigenstates with eigenenergies:

(i) $e_1' : (1\ 0\ 0\ 0)^T$; $E_1' = 15E_0$ (Ans)

(ii) $e_4' : (0\ 0\ 0\ 1)^T$; $E_4' = (3+\frac{3}{100})E_0 = 3.03E_0$ (Ans)

The other 2 eigen energies are obtained by diagonalising the matrix :

$$\frac{1}{E_0}\begin{pmatrix} 3+\frac{3}{100} & \frac{1}{100} \\ \frac{1}{100} & 3 \end{pmatrix}$$

$$\therefore \begin{vmatrix} 3+\frac{3}{100}-\lambda & \frac{1}{100} \\ \frac{1}{100} & 3-\lambda \end{vmatrix} = 0$$

$$\Rightarrow \lambda^2 - \left(6+\frac{3}{100}\right)\lambda + \left(9+\frac{9}{100}-\cancel{10^{-4}}^{\,0}\right) = 0$$

$[10^{-4} \approx 0]$

$$\therefore \lambda = \frac{603 \pm 3}{200}$$

$$\therefore \lambda_1 = \frac{606}{200} = \boxed{3.03E_0 = E_2} \text{ (Ans)} \quad ; \quad \lambda_2 = \frac{600}{200} = \boxed{3E_0 = E_3} \text{ (Ans)}$$

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(1) The unperturbed energy levels are $15E_0$ (non-degenerate) & $3E_0$ (3 fold degenerate).

1st order correct^n for e_1:

$$\Delta E^{(1)}_{e_1} = \langle e_1 | H_p | e_1 \rangle = 0$$

1st order correct^n for 3-fold degenerate states:

They are obtained by diagonalising the perturbing Hamiltonian in the degenerate subspace.

$$\text{so } H_p = \frac{E_0}{100}\begin{pmatrix} 3 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 3 \end{pmatrix} \quad \text{(degenerate subspace)}$$

Let the eigenvalues be λ.

$$\therefore \left|\frac{E_0}{100}\right| \begin{vmatrix} 3-\lambda & 1 & 0 \\ 1 & -\lambda & 0 \\ 0 & 0 & 3-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (3-\lambda)[-\lambda(3-\lambda)] - (3-\lambda) = 0$$

$$\Rightarrow (3-\lambda)[-\lambda(3-\lambda) - 1] = 0$$

$\therefore \lambda_4 = 3$ or $-\lambda(3-\lambda) = 1$

$$\Rightarrow -3\lambda + \lambda^2 - 1 = 0$$

$$\Rightarrow \lambda^2 - 3\lambda - 1 = 0$$

$$\therefore \lambda = \frac{3 \pm \sqrt{9+4}}{2}$$

$$\lambda = \frac{3 \pm \sqrt{13}}{2}$$

$$\boxed{\begin{aligned} \therefore \lambda_2 &= \frac{3+\sqrt{13}}{2} \\ \therefore \lambda_3 &= \frac{3-\sqrt{13}}{2} \end{aligned}}$$

Since the energy would have the factor of $\frac{E_0}{100}$,

$$\Delta E^{(1)}_2 = \frac{E_0}{100} \times \lambda_2 = \frac{3+\sqrt{13}}{200} E_0 \approx \cancel{0.33E_0}\ 0.033 E_0$$

$$\Delta E^{(1)}_3 = \frac{E_0}{100} \times \lambda_3 = \frac{3-\sqrt{13}}{200} E_0 \approx -0.003 E_0$$

$\therefore \boxed{E_{01}^{(1)} = 15E_0}$ (Ans)

$\therefore E_{12}^{(1)} = E_2 + \Delta E_2^{(1)}$ ~~$= 3E_0 + 0.33E_0$~~

$= 3E_0 + 0.033\,E_0 = 3.0$

$\boxed{E_2^{(1)} = 3.033\,E_0}$ (Ans)

$\therefore E_3^{(1)} = E_3 + \Delta E_3^{(1)} = 3E_0 + -0.003E_0$

$\Rightarrow \boxed{E_3^{(1)} = 2.997\,E_0}$ (Ans)

$\therefore E_4^{(1)} = E_4 + \Delta E_4^{(1)}$

$= 3E_0 + \frac{3E_0}{100} = \boxed{3.03\,E_0 = E_4^{(1)}}$ (Ans)

As can be seen, $E_1^{(1)}$ & $E_4^{(1)}$ are ~~perfectly~~ equal to the eigenenergies obtained in part (1). $E_2^{(1)}$ & $E_3^{(1)}$ are also almost equal to the part (1) answers, but slight variations occur since we neglected the factor of 10^{-4}. Overall, ~~all~~ the exact eigenenergies are equal ~~in~~ ~~both~~ to the 1st order corrections.

This happens because H_P does not couple the degenerate $3E_0$ subspace to the $15E_0$ level. The full Hamiltonian splits into independent blocks - 1×1 for e_1, 2×2 for (e_2, e_3) & (1×1) for e_4.